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1.0 User Requirement

Pond Management		
No	Description	Type of relation
1	Record pond info such as: a) Pond ID b) Location of pond c) Size of pond	Attributes
2.	Allow each pond to undergo zero or more pond maintenance conditions during the production cycle.	Multiplicity: Zero to many

Pond's Condition		
No	Description	Type of relation
1	Record pond conditions such as: a) Emptied b) Cleaned c) Refilled d) Reassigned	Attributes

2.	One condition can happen to one or more ponds such as two different ponds carrying out clean activity to clean the fishpond.	Multiplicity: One to many
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Pond's Water Quality		
No	Description	Type of relation
1	Record water quality readings of pond such as: a) Pond ID a) Temperature b) pH c) Dissolved Oxygen Level	Attributes
2.	Each pond can record multiple water quality readings in order to monitor the water quality of the fish ponds.	Multiplicity: One to many
3.	Each water quality readings can be recorded to many ponds as each pond has different water quality readings.	Multiplicity: One to many
4.	Feeding our fish 2 to 3 times a day in each pond to help promote growth.	Multiplicity: One to Many

Batch		
No	Description	Type of relation
1.	Record batch info such as: a) Batch ID b) Pond ID c) StartDate d) EndDate e) Species	Attribute
2.	Each pond can have more than one batch to stock multiple fish species.	Multiplicity: One to many
3.	Each batch must only be assigned to one pond to avoid duplication of the same batch exists in one pond	Multiplicity: One to one

Fish Species		
No	Description	Type of relation
1	Record each species of the fish such as: a) Species b) Size c) Weight d) Age e) Health condition f) Estimated market value	Attribute
2.	Each batch must only consist of one species of fish for focusing on analyzing the growth state of a single fish species.	Multiplicity: One to one
3.	One species of fish can be placed with many batches to investigate the breeding between each of the fish species	Multiplicity: One to many

Fish health status		
No	Description	Type of relation
1	Record health state of fish such as: a) Species b) Health condition c) Mortality rates d) Treatment used	Attribute
2.	Each fish can have many states of health at a time depending on how the feeding process undergoes and the water quality of the batch.	Multiplicity: One to Many
3.	One health status can record for many fish for supporting the health management of the fish.	Multiplicity: One to Many

Staff		
No	Description	Type of relation
1	Record info such as: a) Staff ID b) Staff name c) Role ID d) Contact number	Attribute
2	Each staff member has exactly one role to ensure organizational efficiency, accountability, and clarity.	Multiplicity: One to one
3	The Staff entity is linked to the Feeding Management entity to record feeding activities performed by staff members.	Multiplicity: One to Many
4.	Staff are responsible for performing various daily activities and each staff member will perform multiple activities,	Multiplicity: One to Many
5.	Marketing staff can handle all sales transaction activities.	Multiplicity: One to Many

Role		
No	Description	Type of relation
1	Record info such as: a) Role ID b) Role Name (Marketing/Operational)	Attribute
2.	A single role can be associated with multiple staff members	Multiplicity: One to Many
2.	The role entity shows different roles have different responsibility. For example, an operator will be in charge of feeding, pond maintenance and water management. At the same time, marketing will be in charge of sales and customer management.	Entity

Feeding Management		
No	Description	Type of relation
1	Record info such as: a) Feed ID b) Pond ID c) Batch ID d) Staff ID e) Date f) Time g) Quantity of food h) Cost of food	Attribute
2.	Each feeding activity is performed by exactly one staff member to avoid duplication.	Multiplicity: Many to One
3.	Each feeding activity is restricted to a single pond, ensuring that feeding operations are recorded and managed separately for each pond.	Multiplicity: One to One

Daily Activity Management		
No	Description	Type of relation
1	Capture the following information for each activity record: a) Activity ID b) Pond ID c) Batch ID d) Staff ID e) Activity type ID f) Date of activity g) Time of activity h) Value of activity used i) Cost j) Unit (kWh/Liters)	Attribute
2.	Each activity is restricted to a single activity type, ensuring clear classification and avoiding ambiguity in categorization.	Multiplicity: One to one
3.	An activity may involve multiple equipment items. For example, staff may use 2 equipment at the same time in an activity.	Multiplicity: One to Many

Activity Type		
No	Description	Type of relation
1	Record info such as: a) Activity type ID b) Activity type name - Water replacement - Treatment - Equipment - Electricity consumption - Water consumption	Attribute
2.	A single activity type can be linked to multiple activity records, allowing different activities to be grouped under the same type.	Multiplicity: One to Many

Equipment		
No	Description	Type of relation
1	Record info such: a) Equipment ID b) Equipment name c) Type of Equipment	Attribute
2	One piece of equipment can be used repeatedly, allowing it to support multiple activity records over time	Multiplicity: One to many
3.	The Equipment entity represents farm equipment	Entity

Equipment Usage		
No	Description	Type of relation
1	Record info such as: a) Usage ID b) Equipment ID c) Activity ID d) Time start to used e) Time end to used	Attribute
2.	Each equipment usage record is restricted to a single equipment item, meaning that a usage record cannot involve multiple pieces of equipment.	Multiplicity: Many to One
3.	A single activity can have multiple equipment usage records, allowing different equipment items to be recorded for the same activity.	Multiplicity: Many to One
4..	Equipment Usage records each instance of equipment usage during an activity.	Entity

Customer		
No	Description	Type of relation
1	Record info such as: a) Customer ID b) Customer name c) Customer contact d) Customer address	Attribute
2	One customer can make multiple sales transactions for one fish batch before it is sold out.	Multiplicity: One to many

Sales Transaction		
No	Description	Type of relation
1	Record info such as: a) Sales ID b) Sales quantity c) Price d) Transaction Date e) Payment Record	Attribute
2	Customers can make sales transactions many times.	Multiplicity: One to many
3	Each sales transaction is ordered by one customer.	Multiplicity: Many to one
4	Each sales transaction is considered under one fish batch.	Multiplicity: Many to one
5	One fish batch can be sold through multiple sales transactions until it is fully sold	Multiplicity: One to many

Inventory Item		
No	Description	Type of Relation
1	Record inventory data: A. Item ID B. Item Name C. Quantity in stock D. Unit E. Purchase price F. Expiry date G. Location	Attribute
3	Suppliers can provide many inventory requirements. A single supplier is capable of supplying a diverse range of items.	Multiplicity: One to many
4	Each stock item is mapped back to its specific supplier to ensure quality control and pricing.	Multiplicity: Many to one

Category		
No	Description	Type of Relation
1	Record info: A. Category ID	Attribute
2	Categories organize the inventory hierarchy.	Multiplicity: One to many

Supplier Management		
No	Description	Type of Relation
1	Record supplier details: A. Supplier ID B. Company name C. Contact person D. Contact phone E. Contact email F. Address G. Payment terms	Attribute
3	Suppliers profile maintain multiple product associations. The system tracks all inventory types a specific company can deliver.	Multiplicity: One to many
4	Inventory items support multi-vendor sourcing. Critical items can be procured from several different suppliers to mitigate the risk of stock shortages.	Multiplicity: Many to many

Purchase Order		
No	Description	Type of Relation
1	Record purchase order: A. Purchase Order ID B. Purchase Order Date C. Total Cost D. Status (Received/Pending) E. Payment Terms	Attribute
3	A single transaction document can include a variety of	Multiplicity:

	different supplies needed for the company's daily operation.	One to many
4	Each purchase order is issued to exactly one supplier to maintain clear financial and legal accountability.	Multiplicity: Many to one
5	The same product is frequently requested across many different purchase orders within a year. This historical data is used to analyze consumption rates and price fluctuations.	Multiplicity: One to many

Purchase Order Detail		
No	Description	Type of Relation
1	Record purchase order detail: A. Purchase Order Detail ID B. Quantity Ordered C. Unit Price At Purchase	Attribute
2	The same product is frequently requested across many different purchase orders within a year. This historical data is used to analyze consumption rates and price fluctuations.	Multiplicity: One to many

Reporting & Monitoring		
No	Description	Type of Relation
1	Record cost consolidation data: A. Cost ID B. Cost Type C. Amount D. Date E. Activity Type	Attribute
3	Frequent feeding sessions create a stream of data entries that are logged against specific ponds. These individual events are later aggregated to monitor the total cost of fish growth per batch.	Multiplicity: One to many
4	Every cost recorded in the system must be identified by a specific category. This classification allows management to pinpoint which activities are driving the highest expenses.	Multiplicity: Many to one

Profit Record		
No	Description	Type of Relation
1	Record production & financial metrics: A. Total Production Cost B. Total Sales Revenue C. Mortality Rates/Losses D. Profit per Batch E. Efficiency Indicators(KPIs)	Attributes
2	Financial summary report generated for each batch upon completion of all sales transactions	Multiplicity: One to one

2.0 Assumption

1. Unique Identification

- Each major entity in the system such as ponds, fish batches, employees, inventory items, suppliers, and customers is assumed to have a unique identification number to ensure accurate tracking and database relationships.

2. Single Pond Assignment for Batch

- A fish batch is assumed to be assigned to only one pond at a time during a production cycle. However, a pond may contain multiple batches depending on the farm's stocking strategy.

3. One species of fish for one pond

- Assume only one species of fish is stored in one batch at a time in order to record the detailed information for each of the species of fish accurately.

4. Payment Method

- The system assumes that all sales payments are made via cash or bank transfer.
- No online payment gateway integration such as e-wallets or credit card platforms is required.
- Payments are recorded immediately in the system once the transaction is completed

5. Fish Mortality

- Fish deaths are recorded daily and linked to their specific batch and pond.
- The system assumes staff will inspect fish daily to update health and mortality records.
- No disease diagnosis module is included; only health status categories (Healthy, Sick, Dead) are tracked.
- Fish that die from unexpected causes (disease, accident, or environmental factors) are recorded under mortality in health status.

6. Inventory and Supplies

- Inventory tracking assumes that all stock is purchased from verified suppliers.
- The system does not account for manual stock loss or theft — stock updates occur only when purchase or usage is recorded.
- Expiry dates are monitored only for perishable items such as feed, medicine and chemicals.

7. System Access and Permissions

- Staff can only enter data related to their role.
- Operational staff enter feeding, pond maintenance, and water quality records.
- Marketing staff enter sales and customer details.

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